Options for Achieving Interoperability

Explore different ways for achieving interoperability as well as their security tradeoffs.

DEC
12th
3:30pm PT
8:30am KST

Charlie Noyes
Partner
Paradigm

INTERCHAIN
CONVERSATIONS

conversations.cosmos.network

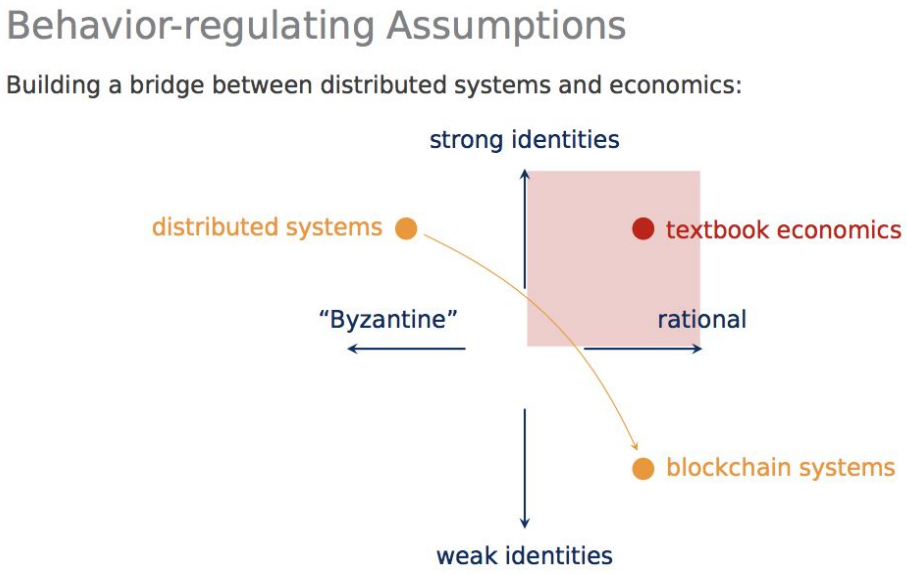
CØSMOS

The Security of Interoperability

GO WATCH THE PRESENTATION HERE:
https://youtu.be/QdnQUIPeIzY?t=54

The Blockchain Context

- A blockchain enables agreement on some computation between participants who are:
  - Pseudonymous
  - Rational (adversarial)
  - Able to corrupt any “trusted” 3rd party
- *The* question: how does a blockchain incentivize consensus participants to follow intended behavior?



Three Axioms of Security

1. Regularity is Safety
   -
   -
2. Users are Parasites
   -
   -
   -
3. Dimensionality is a Curse
   -
   -

Regularity is Security

- Two key observations from “On the Instability of Bitcoin Without the Block Reward”:
  - Only stable and predictable miner revenue streams, such as inflationary block rewards, create an incentive for miners to securely extend the chain (add to the “security budget”)
  - Irregular and unpredictable miner revenue streams, including demand-based transaction fees, actually create an incentive for miners to 51% attack and reorg the chain (*detract* from the security budget)
- Practically, this means that the value of a blockchain’s native asset must be able to incentivize consensus stability at sustainable levels of issuance (re-distribution)

Three Axioms of Security

1. Regularity is Security
   - Only stable and predictable miner revenue streams can encourage consensus stability
   - Irregular miner revenues, such as non-deterministic txfees, encourage consensus **in**stability
2. Users are Parasites
   -
   -
   -
3. Dimensionality is a Curse
   -
   -

Users are Parasites

- Behavior occurring at a blockchain's application layer can create incentives for consensus participants which are not contemplated in the core protocol
- Rooted in "Flash Boys 2.0," which coined "Miner Extractable Value" (MEV) to describe these extra-protocol incentives
  - One of the more nuanced ideas here, so we'll take a brief detour to explain it

Flash Boys 2.0: Frontrunning, Transaction Reordering, and Consensus Instability in Decentralized Exchanges

Philip Daian, *Cornell Tech*, phil@cs.cornell.edu
Steven Goldfeder, *Cornell Tech*, goldfeder@cornell.edu
Tyler Kell, *Cornell Tech*, sk3259@cornell.edu
Yunqi Li, *UIUC*, yunqil3@illinois.edu
Xueyuan Zhao, *CMU*, xyzhao@cmu.edu
Iddo Bentov, *Cornell Tech*, ib327@cornell.edu
Lorenz Breidenbach, *ETH Zürich*, lorenz.breidenbach@inf.ethz.ch
Ari Juels, *Cornell Tech*, juels@cornell.edu

v1 [cs.CR] 10 Apr 2019

***Abstract*—Blockchains, and specifically smart contracts, have promised to create fair and transparent trading ecosystems.**

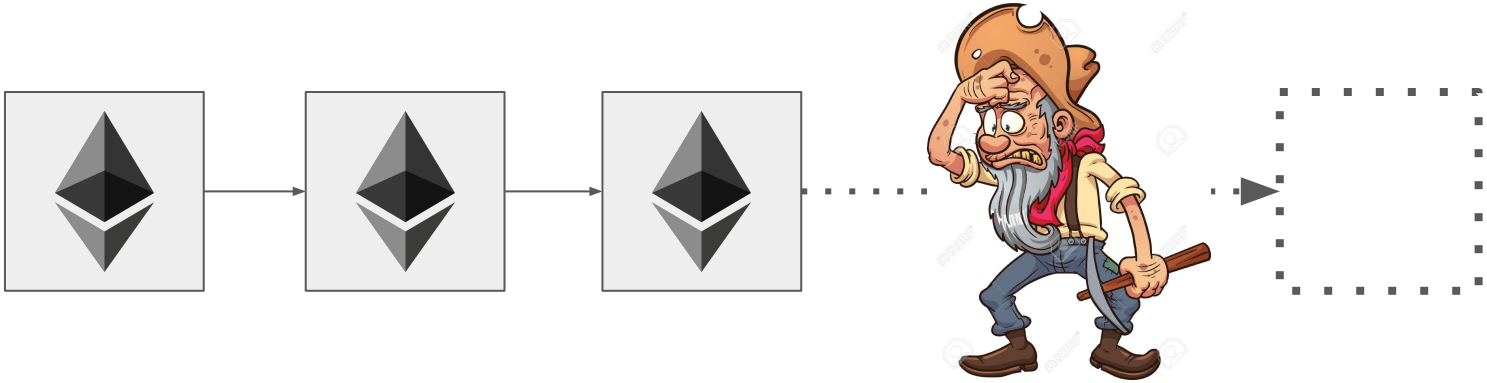
Unfortunately, we show that this promise has not been met. We document and quantify the widespread and rising deployment of *arbitrage bots* in blockchain systems, specifically in *decentralized exchanges* (or "DEXes"). Like high-frequency traders on Wall Street, these bots exploit inefficiencies in DEXes, paying high transaction fees and optimizing network latency to frontrun, i.e., anticipate and exploit, ordinary users' DEX trades.

We study the breadth of DEX arbitrage bots in a subset of transactions that yield quantifiable revenue to these bots. We also study bots' profit-making strategies, with a focus on blockchain-specific elements. We observe bots engage in what we call *priority*

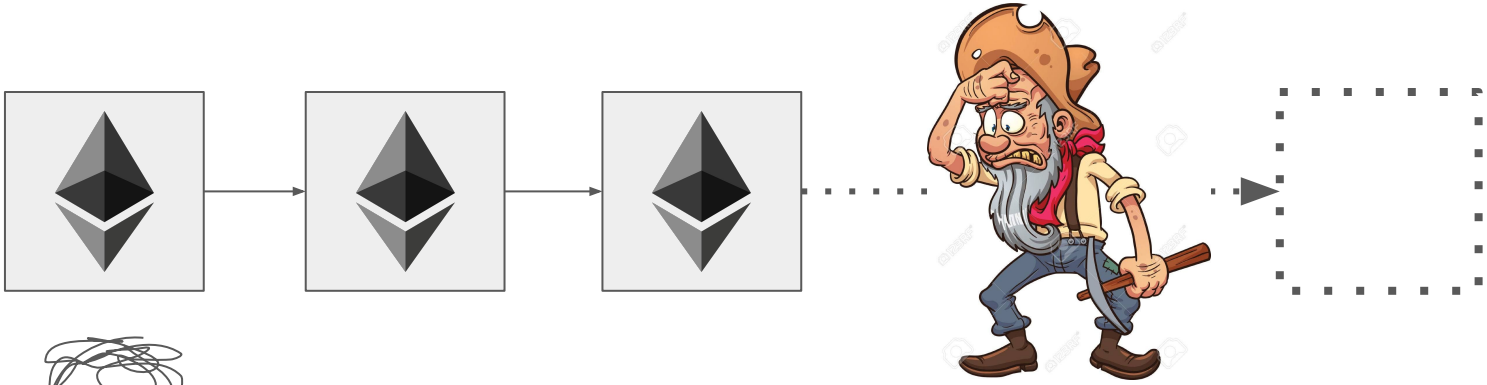
decentralized exchange (or "DEXes").[1] In a DEX, a smart contract (a program executing on a blockchain) or other form of peer-to-peer network executes exchange functionality.

At first glance, decentralized exchanges seem ideally designed. They appear to provide effective price discovery and fair trading, while doing away with the drawbacks of centralized exchanges. Trades are atomically executed by a smart contract and visible on the Ethereum blockchain, providing the appearance of transparency. Funds cannot be stolen by the exchange operator, because their custody and exchange logic is processed and guaranteed by the smart contract.

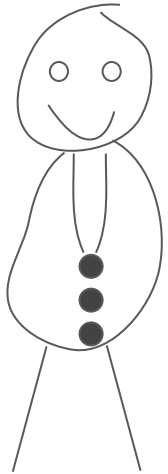
Flash Boys 2.0



Flash Boys 2.0

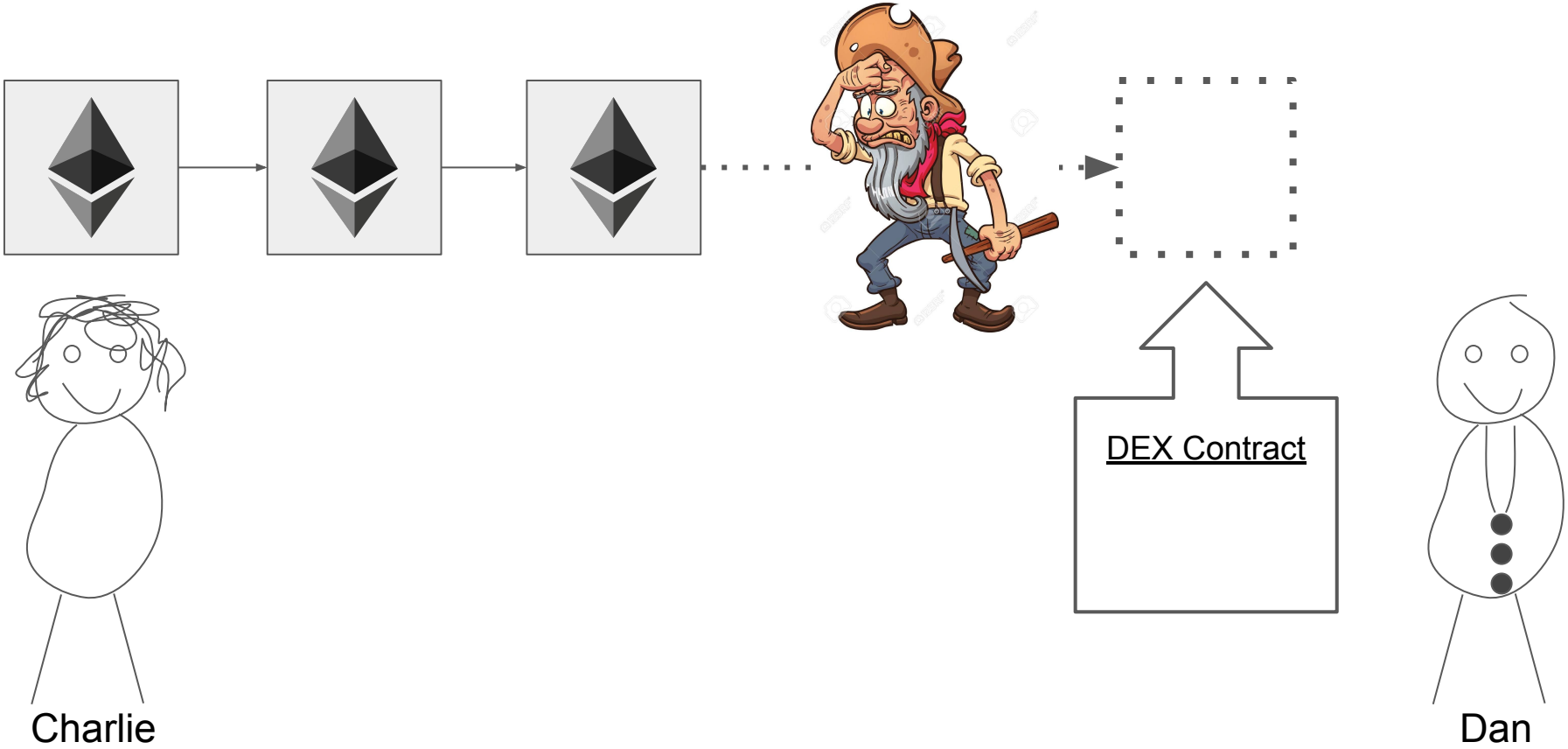


Charlie

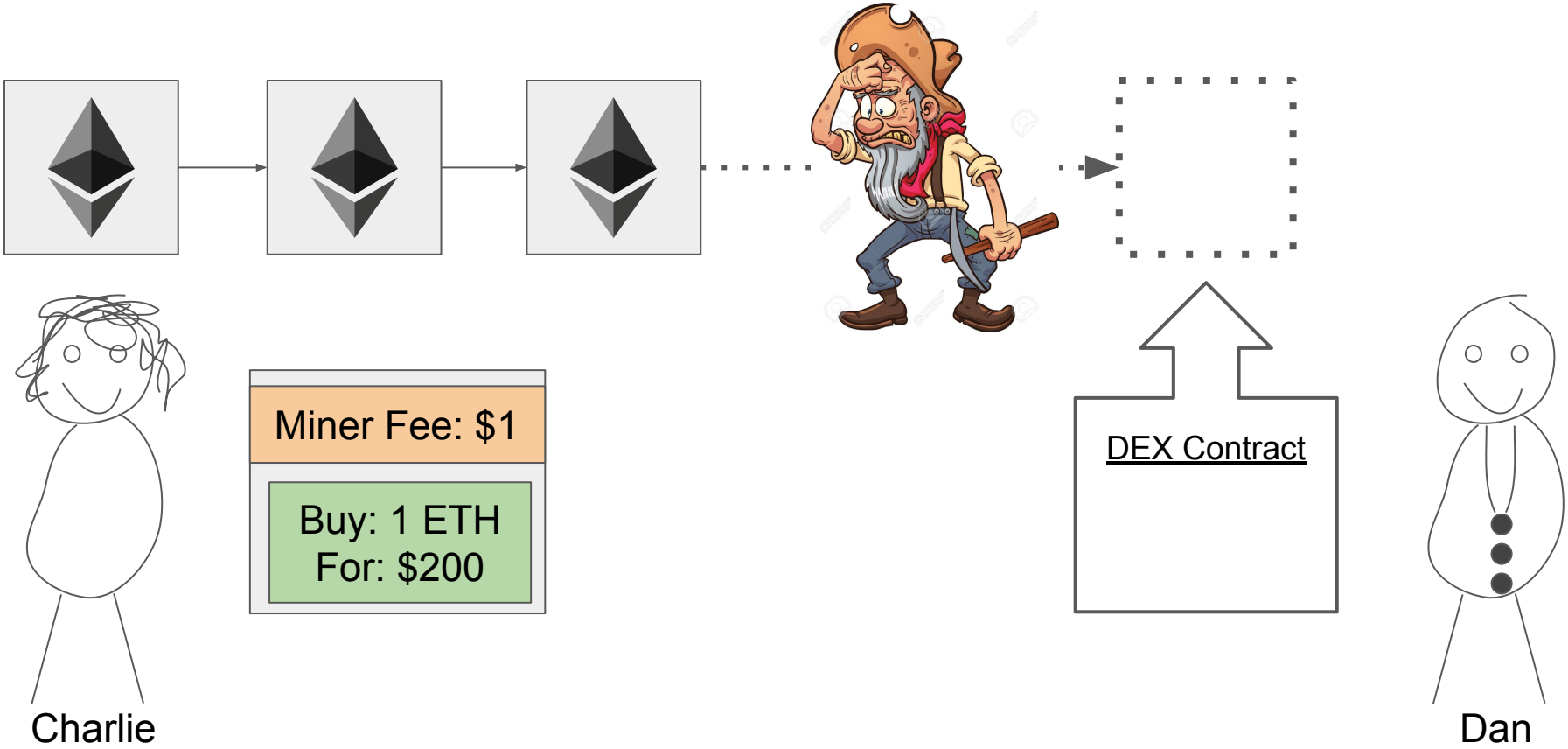


Dan

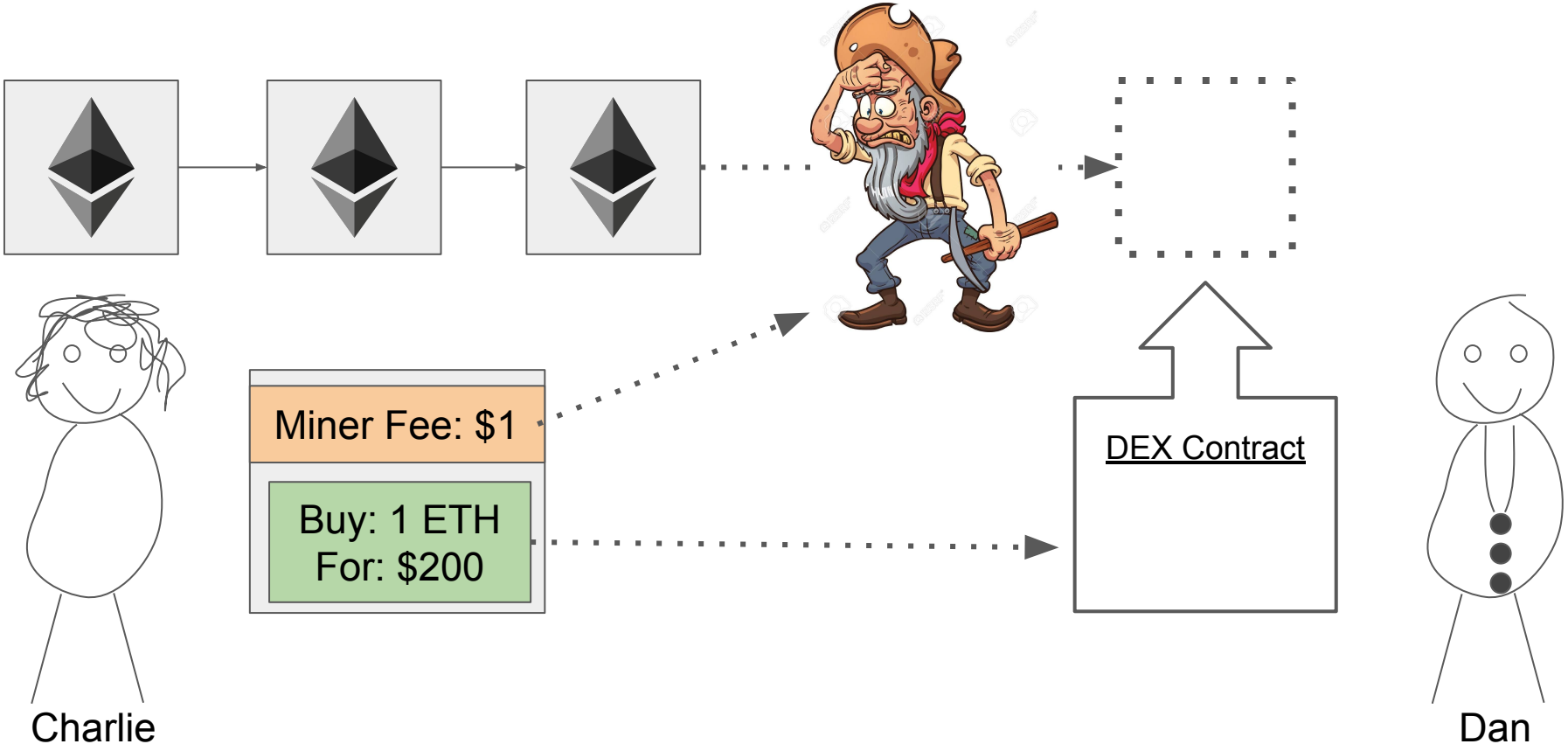
Flash Boys 2.0



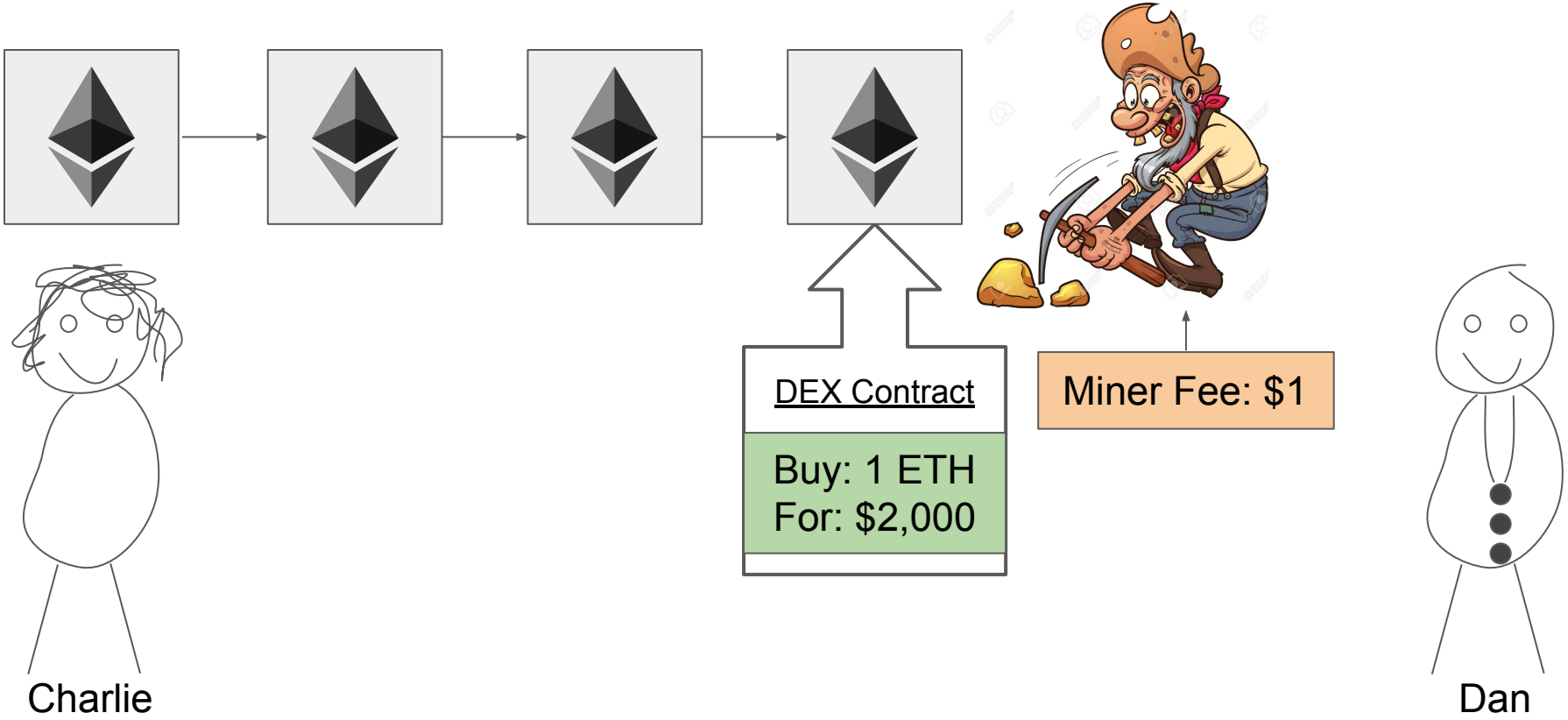
Flash Boys 2.0



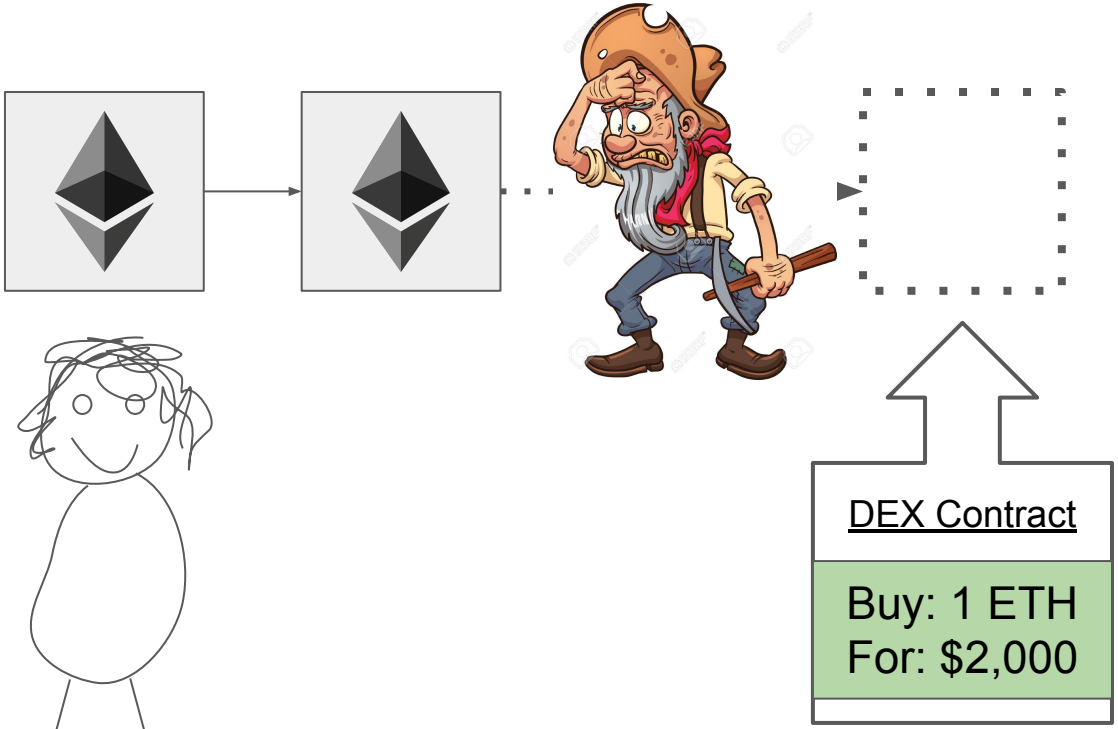
Flash Boys 2.0



Flash Boys 2.0



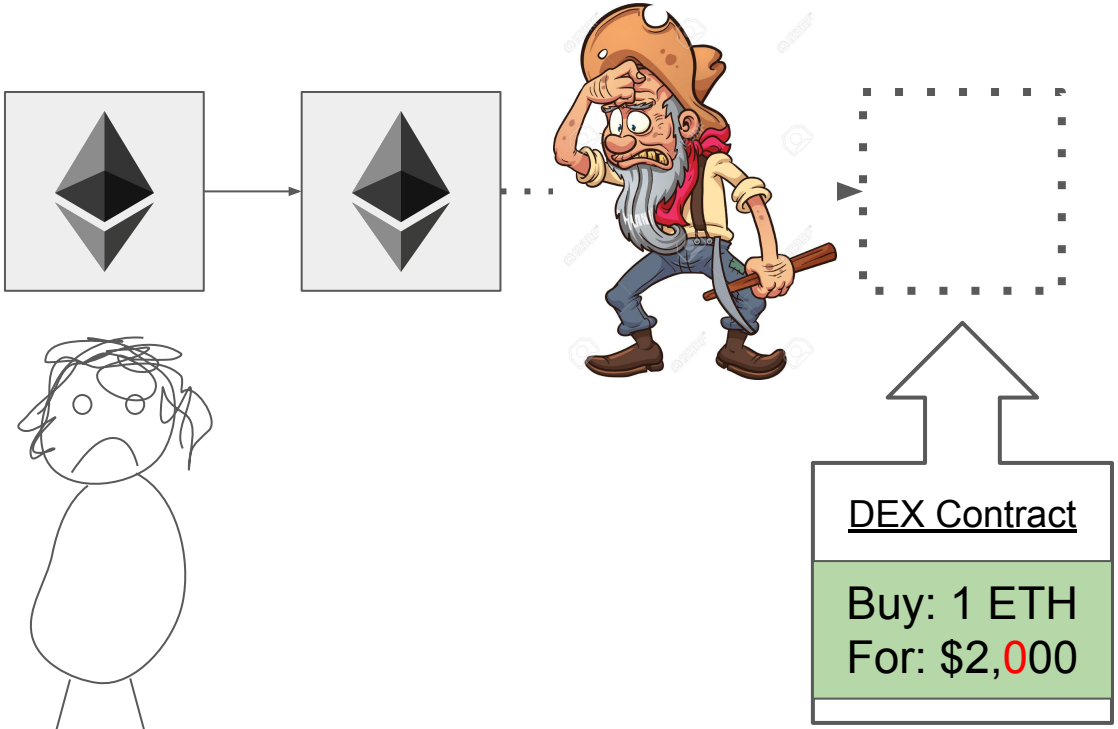
Flash Boys 2.0



Charlie

Dan

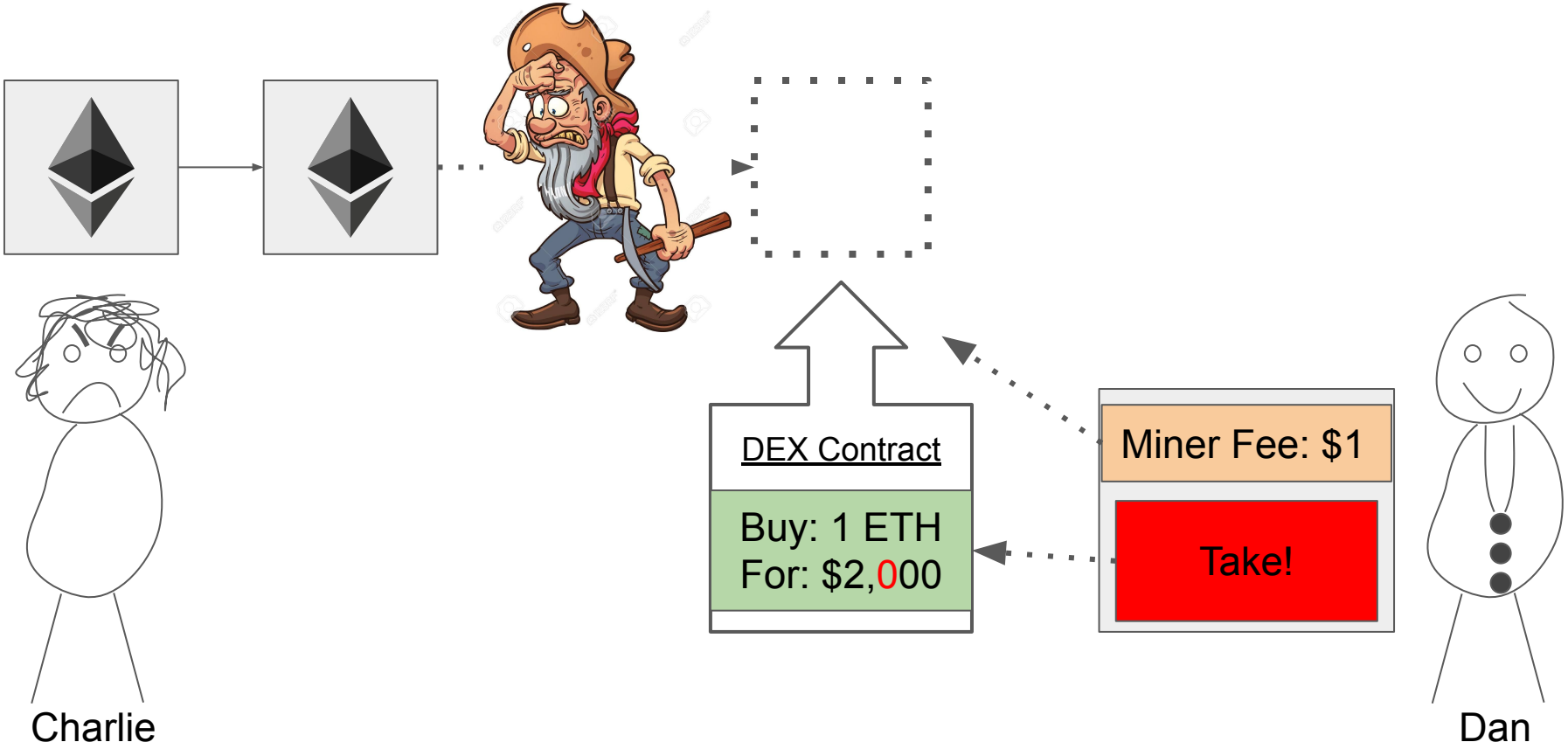
Flash Boys 2.0



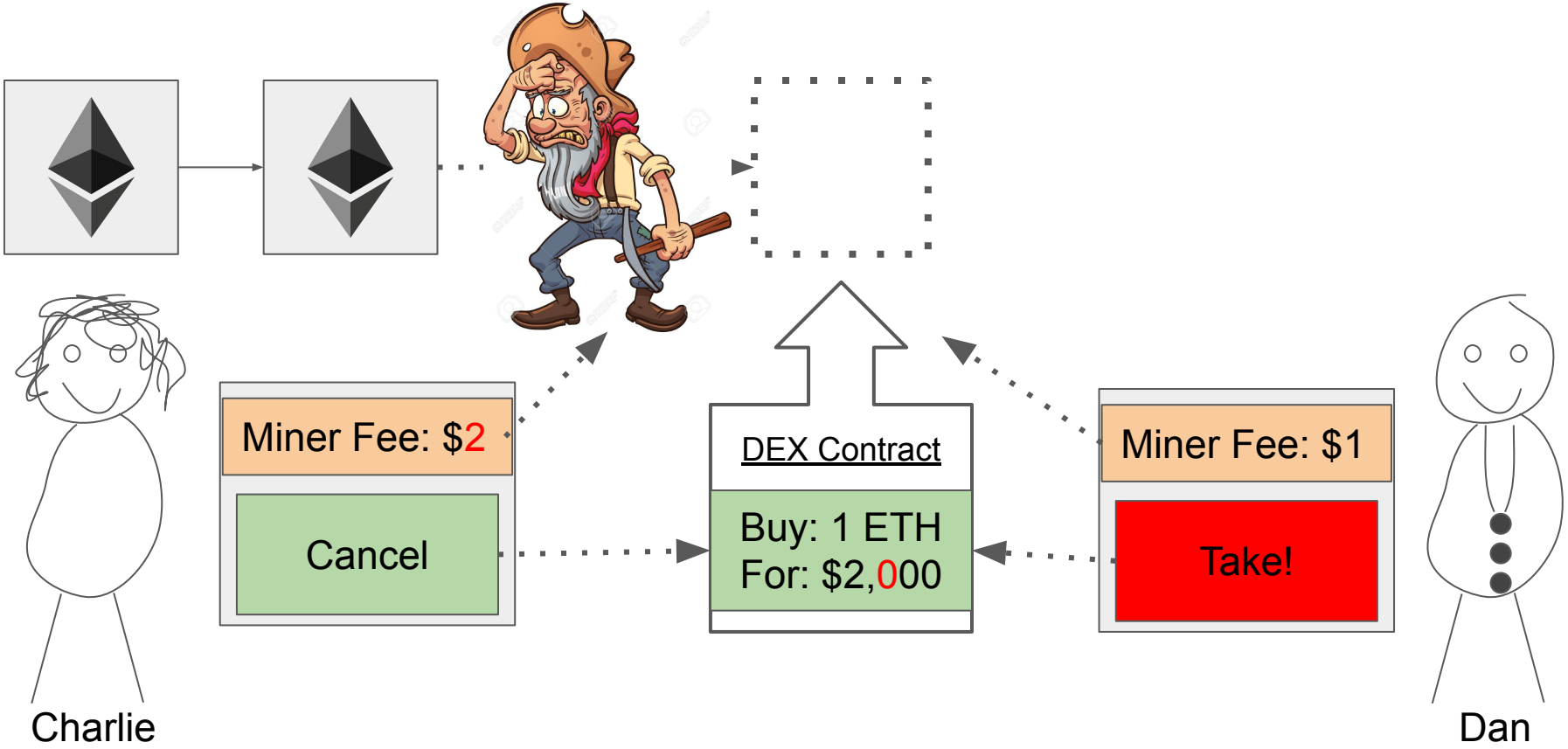
Charlie

Dan

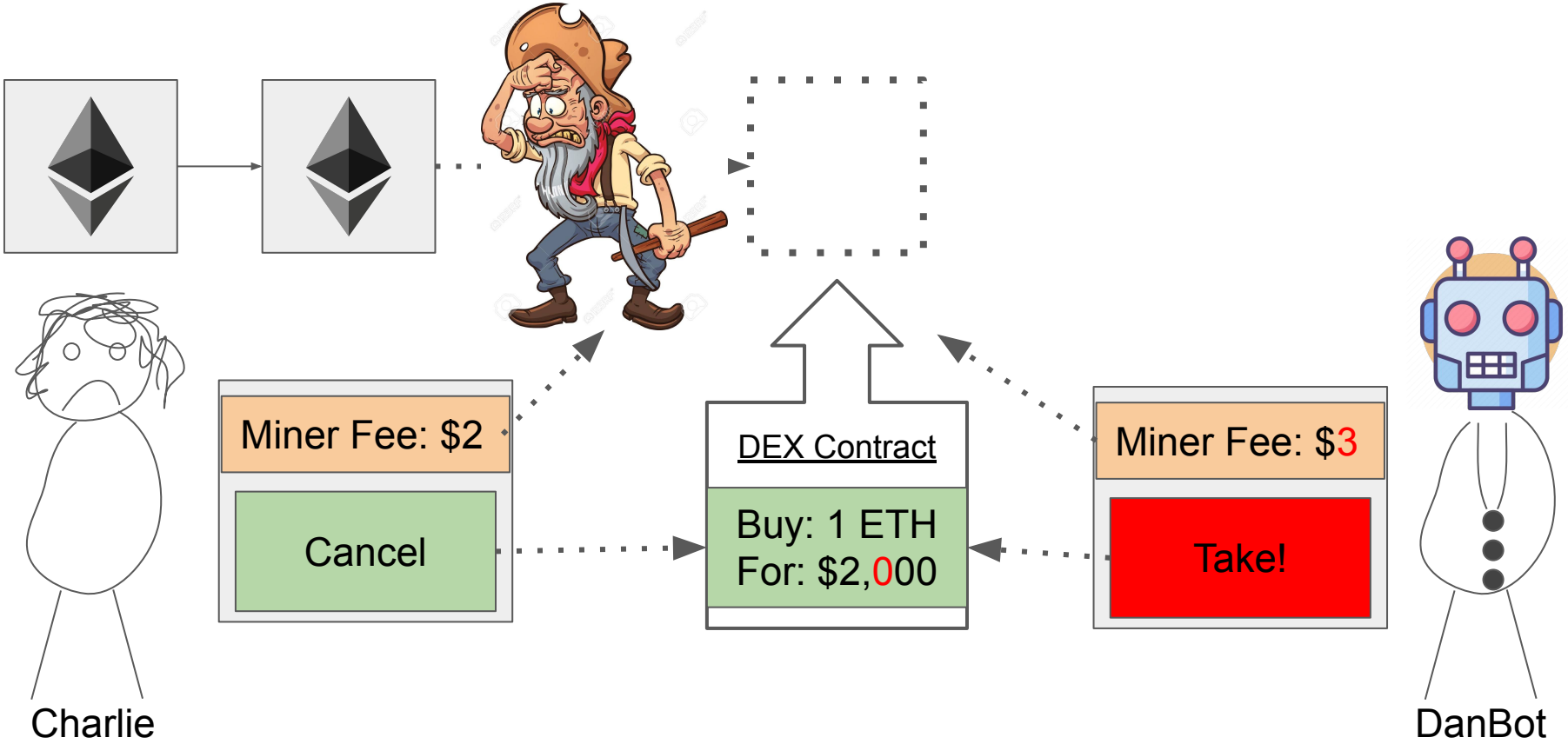
Flash Boys 2.0



Flash Boys 2.0



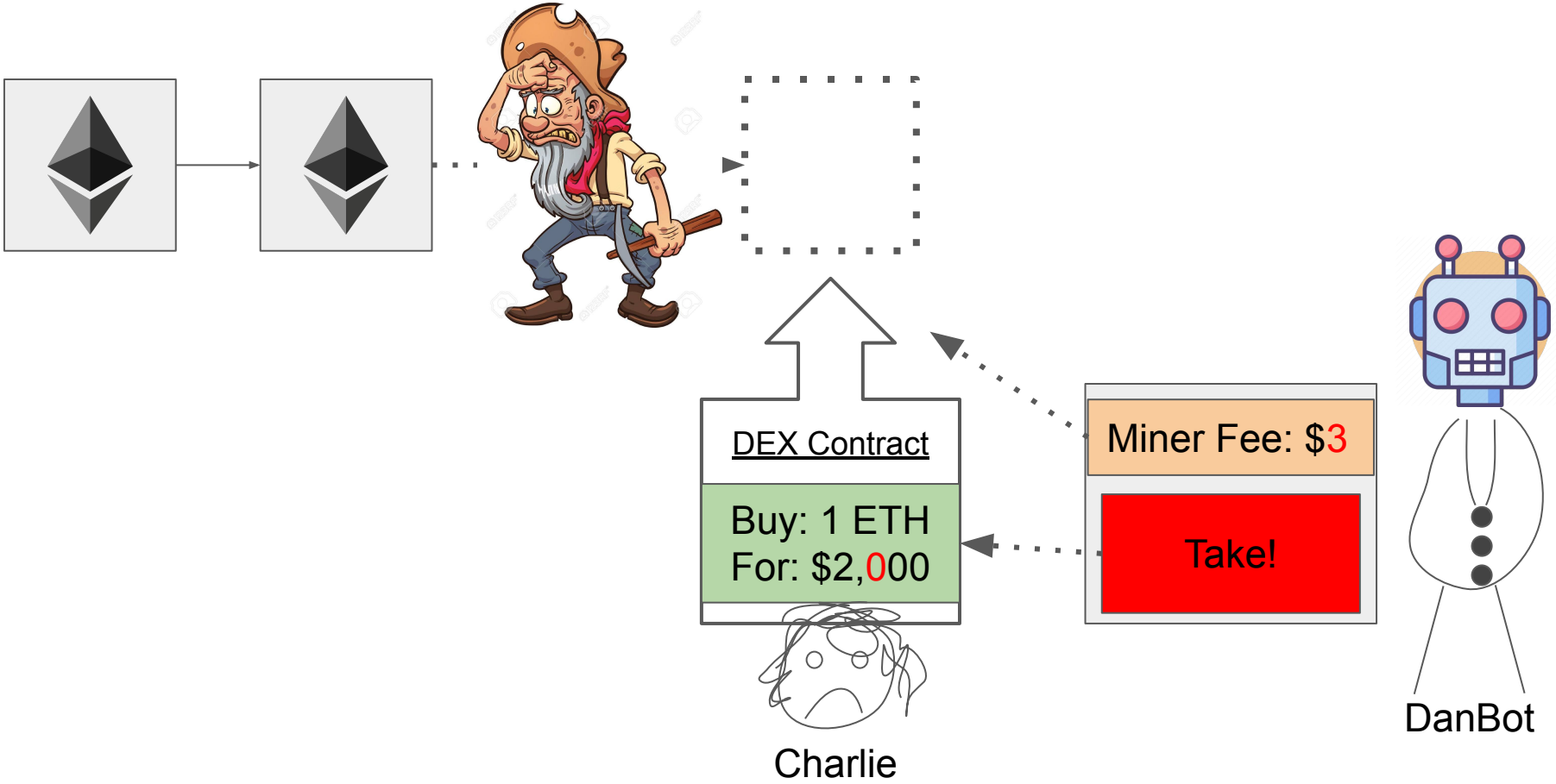
Flash Boys 2.0



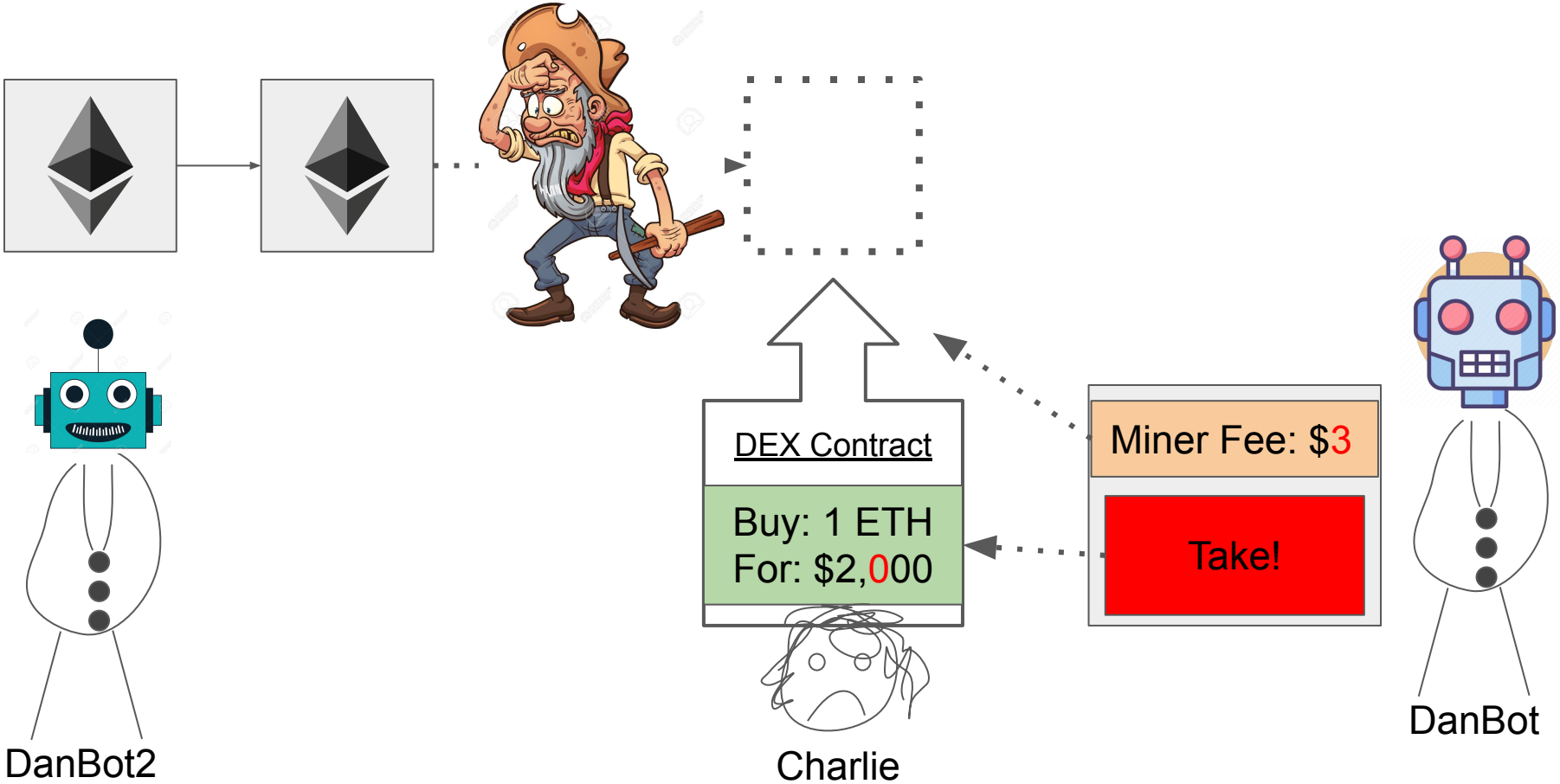
Flash Boys 2.0

Intuition: DanBot pays extra fee to miner to front-run Charlie!

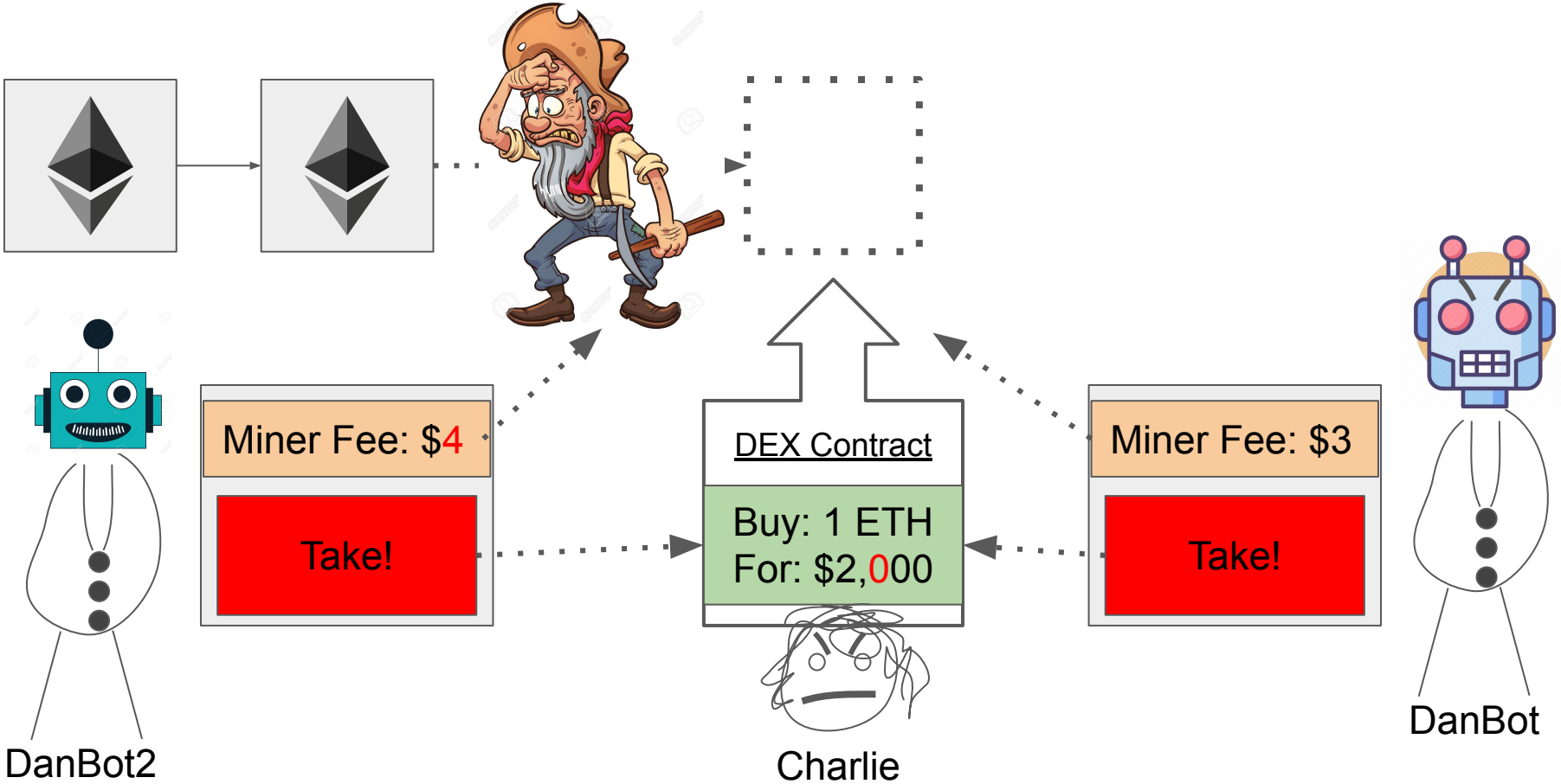
Flash Boys 2.0



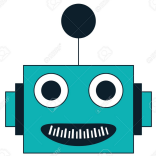
Flash Boys 2.0



Flash Boys 2.0



Flash Boys 2.0



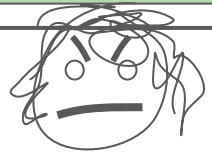
DanBot2

Miner Fee: $4

Take!

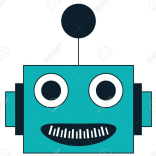
DEX Contract

Buy: 1 ETH
For: $2,000



Charlie

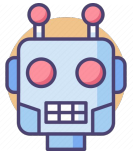
Flash Boys 2.0



DanBot2

Miner Fee: $4

Take!



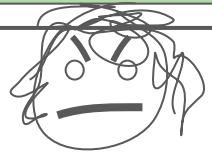
DanBot

Miner Fee: $5

Take!

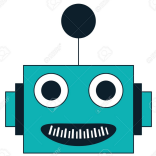
DEX Contract

Buy: 1 ETH
For: $2,000



Charlie

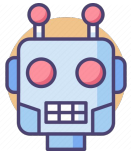
Flash Boys 2.0



DanBot2

Miner Fee: $4

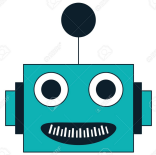
Take!



DanBot

Miner Fee: $5

Take!



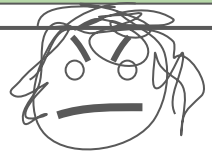
DanBot2

Miner Fee: $6

Take!

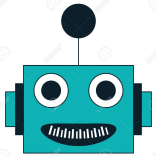
DEX Contract

Buy: 1 ETH
For: $2,000



Charlie

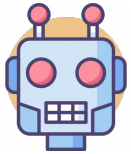
Flash Boys 2.0



DanBot2

Miner Fee: $4

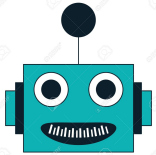
Take!



DanBot

Miner Fee: $5

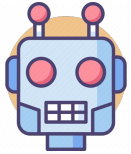
Take!



DanBot2

Miner Fee: $6

Take!



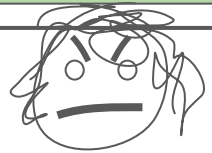
DanBot

Miner Fee: $7

Take!

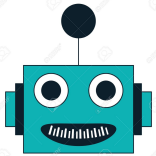
DEX Contract

Buy: 1 ETH
For: $2,000



Charlie

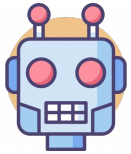
Flash Boys 2.0



DanBot2

Miner Fee: $4

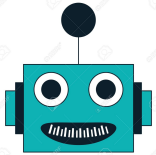
Take!



DanBot

Miner Fee: $5

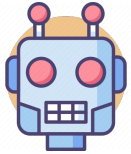
Take!



DanBot2

Miner Fee: $6

Take!



DanBot

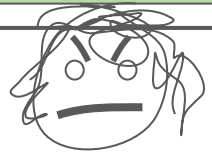
Miner Fee: $7

Take!

...

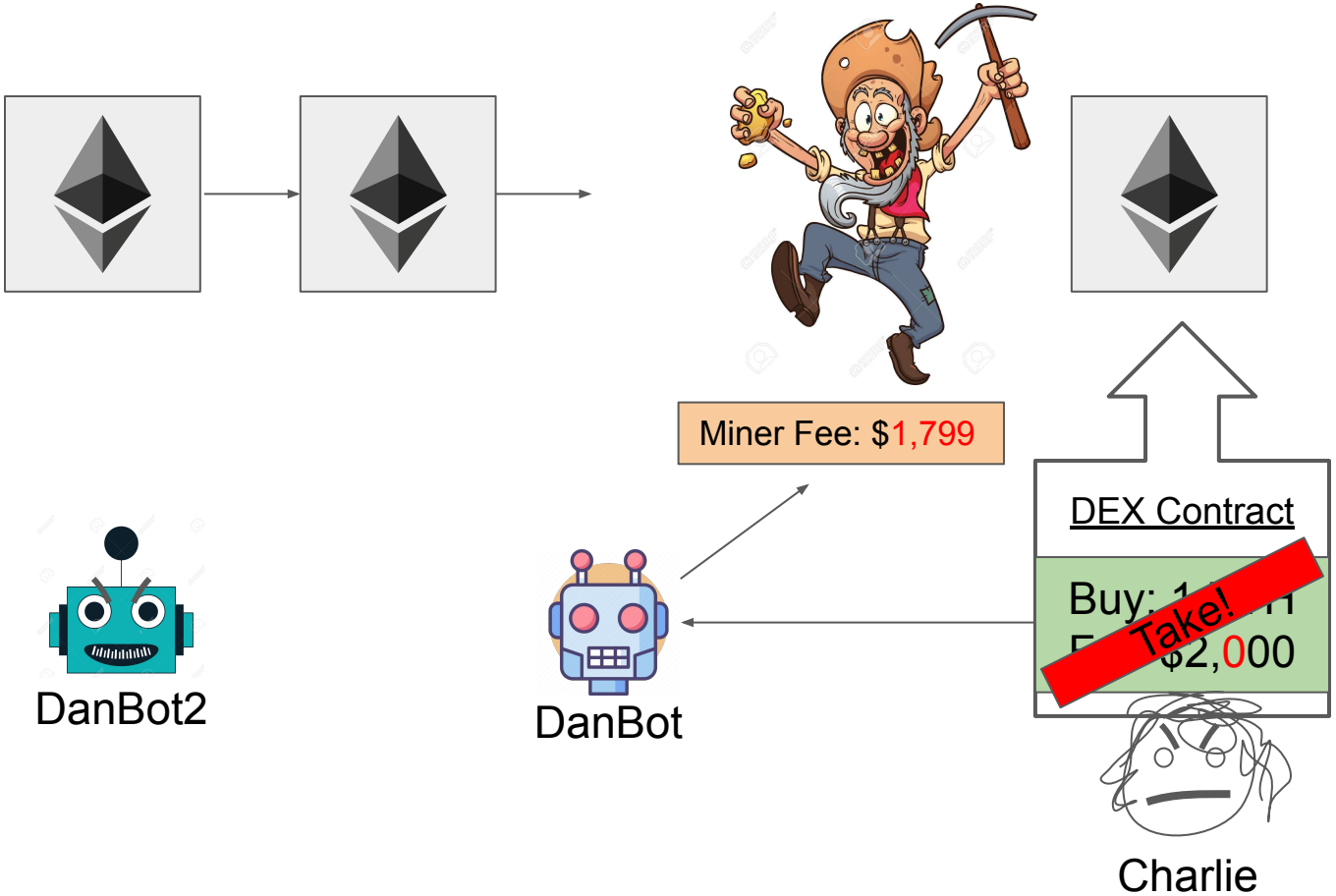
DEX Contract

Buy: 1 ETH
For: $2,000



Charlie

Flash Boys 2.0



Flash Boys 2.0

- This DEX scenario was an example of MEV able to be captured over the span of a single block -- this can happen in one of two ways:
  - Like we just saw: non-miner arbitrageurs competing to capture the MEV play out the gas (txfee) auction game. There is “no free lunch” in theoretical equilibrium, so we must assume that the txfees will approach the value of the arbitrage.
  - Alternatively: a miner simply censors competing transactions to capture it themselves. From the perspective of the miners’ incentives, this is theoretically equivalent to receiving the txfee.
- MEV can also exist across the span of multiple blocks: opportunities that are only able to be captured via a reorg
  - E.g., opening a leveraged position and reorging it out of existence if it becomes unprofitable.

Flash Boys 2.0

- Single-block capturable MEV also creates an incentive to reorg the chain, in the same way as “simple” transaction fees did earlier
  - Of course, we can now view those txfees as just one particular form of MEV
- Therefore: MEV can exist on any timescale, and all MEV contributes to an incentive to reorg the chain cumulative across *all* timescales
- Yes, this is getting disturbing...

Users are Parasites

- Can we construct a different method of transaction ordering (fee scheme) which prevents MEV from being externalized as “irregular miner revenue?”
- No, for three reasons:
  a. The MEV contained in any given transaction is *a priori* unable to be known by the blockchain.
     - This is obviously the case in Ethereum-like blockchains.
     - It is also true of Bitcoin, albeit less obviously. Quoting Hasu:
       - “The Bitcoin system is highly optimized for the transfer of bitcoins, but there are limits to the extent to which the storage of arbitrary data can be discouraged. Because this arbitrary data can represent unbounded value outside of the Bitcoin network, [...] it injects potentially limitless MEV and adds incentive to attack the chain.”
  b. …
  c. ...

Users are Parasites

- Can we construct a different method of transaction ordering (fee scheme) which prevents MEV from being externalized as “irregular miner revenue?”
- No, for three reasons:
  a. The MEV contained in any given transaction is *a priori* unable to be known by the blockchain.
     - This is obviously the case in Ethereum-like blockchains.
     - It is also true of Bitcoin, albeit less obviously. Quoting Hasu:
       - [...]
  b. A user is willing to pay as much for transaction ordering as the MEV profit opportunity admitted (no free lunch). If there is no way to express this demand in-protocol, miners and users will collude in off-protocol markets. Therefore, txfees must *be able to be* proportional to MEV.
     - This is why Ethereum’s EIP-1559 includes a “tip auction" in addition to the flat BASEFEE; any transaction that contains MEV will be competed over in a tip auction.
  c. Txfees which are proportional to MEV are themselves MEV.

Three Axioms of Security

1. Regularity is Security
   - Only stable and predictable miner revenue streams can encourage consensus stability
   - Irregular miner revenues, such as non-deterministic txfees, encourage consensus *instability*
2. Users are Parasites
   - User behavior occurring at a blockchain's application layer can create incentives for consensus participants which are not contemplated in the core protocol ("MEV")
   - All forms of MEV are inevitably *able* to be externalized as "irregular miner revenues," and therefore encourage consensus instability
   - The quantity and magnitude of all *hypothetically possible* MEV vectors is always infinite for all blockchains
3. Dimensionality is a Curse
   -
   -

Dimensionality is a Curse

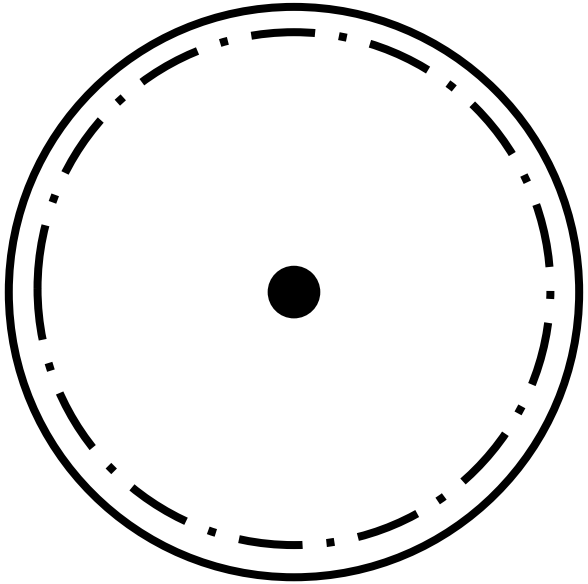
- Although the potential for perverse MEV incentives is necessarily infinite in any blockchain context, the rate at which they accumulate can differ
- Hypothesis: the more levers you have to play with, the more ways there are for MEV to manifest itself, and (in general) the faster it will accumulate
  - In Bitcoin, only irregular transaction fees, HTLC's, and colored coins obviously come to mind. There is some pathological future in which the value of colored coins exceeds the value of BTC (and thus the miner fee stream) by a factor of 1,000,000x, but it's hard to imagine.
  - Ethereum is already empirically seeing behavior driven by an uncountable number of MEV vectors, nobody feels they can see the lower bound of their magnitude, and there is a uniform consensus opinion that MEV incentives are likely to continue increasing

Dimensionality is a Curse

- We call this a blockchain's "dimensionality" -- the abstract degrees of freedom within which its asymptotic incentive behavior may independently vary
- Hypothesis: limiting dimensionality requires hard in-protocol constraints on allowed application layer behavior
  - If it were discovered Lightning channels created untenable MEV and threatened Bitcoin's consensus stability, we could disable their opcodes
  - If it were discovered that some application pattern (e.g. DEX's) posed similar risks to Ethereum, it would be impossible to preclude all possible implementations of those behaviors at the level of the EVM
    - Individual implementations could be forked off, but we could not prevent the general behavior without permissioning contract deployment or constraining the EVM to such a degree that Ethereum would no longer be a "smart contract" blockchain

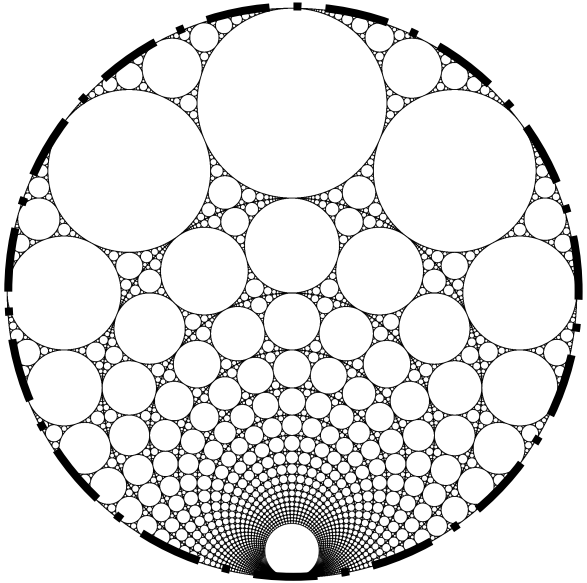
Dimensionality is a Curse

Bitcoin



Dimensionality: Minimal

Ethereum



Dimensionality: Arbitrarily High

Dimensionality is a Curse

- There are security-additive incentives other than inflationary block rewards:
  - EIP-1559's BASEFEE burns
    - Although not EIP-1559 tip auction revenue, which is MEV
  - State rents
  - Probably many other potential constructions
- These incentives are lower-dimensional than MEV, and therefore cannot be guaranteed to scale proportionally to or asymptotically dominate MEV
  - EIP-1559 BASEFEEs: in the absolute # of transactions made on Ethereum
    - Pathological counterex. - only a single transaction is made each day, which creates a $1T frontrunning opportunity
  - State rents: in the demand for Ethereum state capacity
    - Pathological counterex. -

Three Axioms of Security

1. Regularity is Security
   - Only stable and predictable miner revenue streams can encourage consensus stability
   - Irregular miner revenues, such as non-deterministic txfees, encourage consensus *instability*
2. Users are Parasites
   - User behavior occurring at a blockchain's application layer can create incentives for consensus participants which are not contemplated in the core protocol ("MEV")
   - All forms of MEV are inevitably *able* to be externalized as "irregular miner revenues," and therefore encourage consensus instability
   - The quantity and magnitude of all *hypothetically possible* MEV vectors is always infinite for all blockchains
3. Dimensionality is a Curse
   - MEV will (generally) increase more quickly the more dimensions there are for it to grow in, and the only way to limit dimensionality is via constraining application-layer behavior in-protocol
   - Security-additive incentive constructions are lower-dimensional than MEV and cannot be *guaranteed* to scale proportionally to or asymptotically dominate MEV

Extrapolating out

- If Bitcoin doesn't work, all of our geese are probably cooked
  - Modulo block rewards going to zero in the extreme long-term
- Ethereum has basically two futures:
  - It works, due to some combination of:
    - ETH is simply so valuable that it is able to socialize the cost of securing Ethereum essentially invariant to the scale of MEV incentives
    - Non-inflationary stable revenues grow faster than MEV incentives (they can't be *guaranteed to*, but that doesn't mean they *can't*)
      - EIP-1559's burned BASEFEEs, state rent, etc.
    - Users dislike getting frontrun and as MEV becomes more obviously harmful, application developers are incented to adopt design patterns which minimize MEV exposure
  - MEV comes to dominate the value of miner security incentives
    - This probably implies that the security of any Ethereum-like system (arbitrary application behavior) is pathologically degenerate
    - This would be quite disappointing!

The Security of Interoperability

(I promised we would get there…)

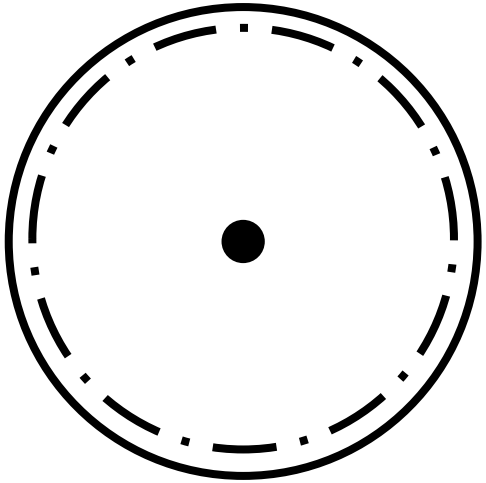
Interoperability: Two Poles

- Bitcoin: a *Black Domain*
  - Zero interoperability - has no understanding of external consensus messages

- Ethereum: a *Dark Forest*
  - High interoperability - nearly frictionless as all (arbitrary) behavior is homogeneously secured
  - Uniformly sharing security meants the incentives of all applications are externalized on all others, and without provisioning deployment you cannot preclude specific behaviors
  - May be pathologically degenerate

Interoperability: Two Poles

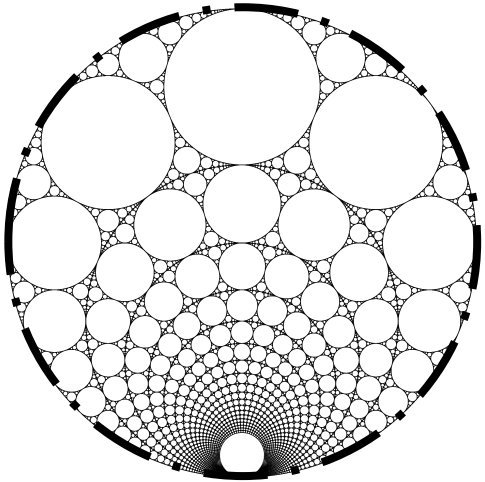
Bitcoin



Dimensionality: Minimal

Interoperability: Minimal

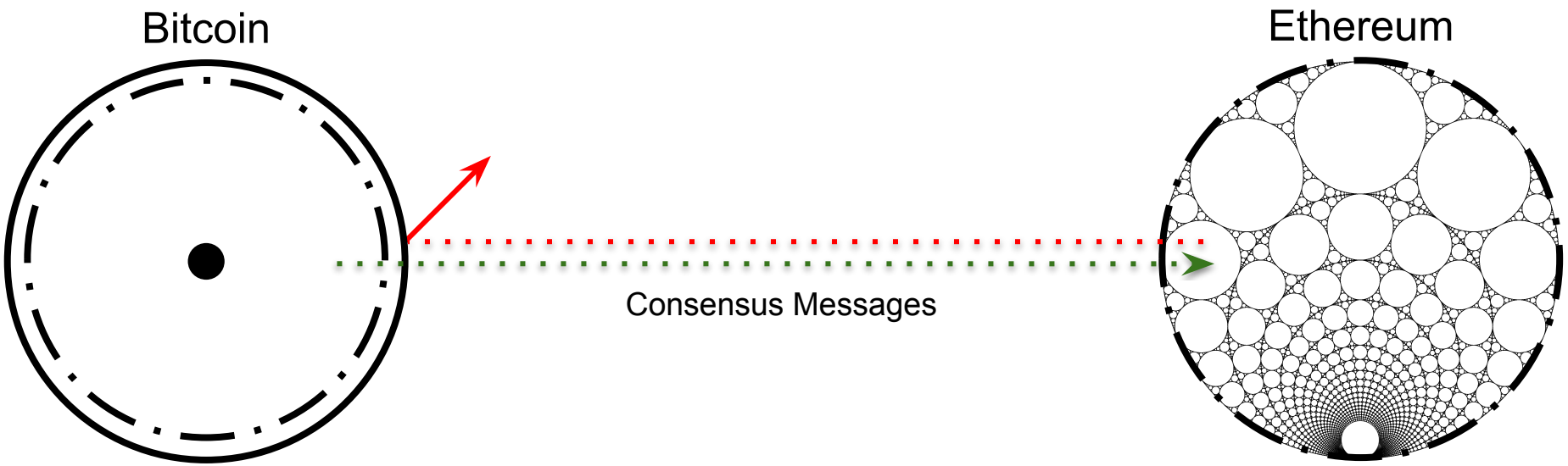
Ethereum



Dimensionality: Arbitrarily High

Interoperability: High

Interoperability: Two Poles



Dimensionality: Minimal

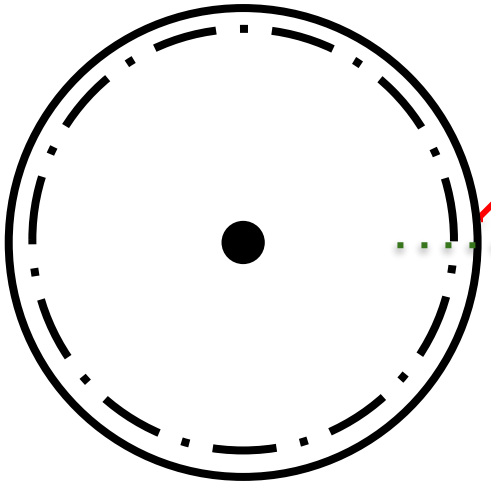
Interoperability: Minimal

Dimensionality: Arbitrarily High

Interoperability: High

Interoperability: Two Poles

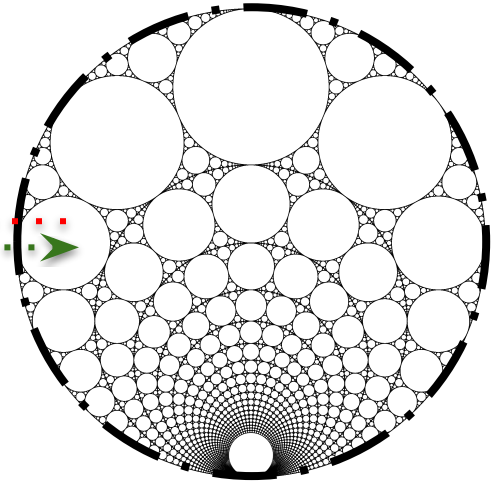
Bitcoin



(👇 why 2way BTC pegs are tough)

Consensus Messages

Ethereum



Dimensionality: Minimal

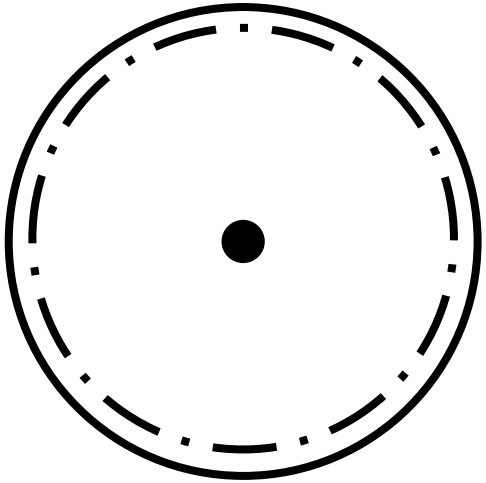
Interoperability: Minimal

Dimensionality: Arbitrarily High

Interoperability: High

Interoperability: The Third Door

Bitcoin



Dimensionality: Minimal

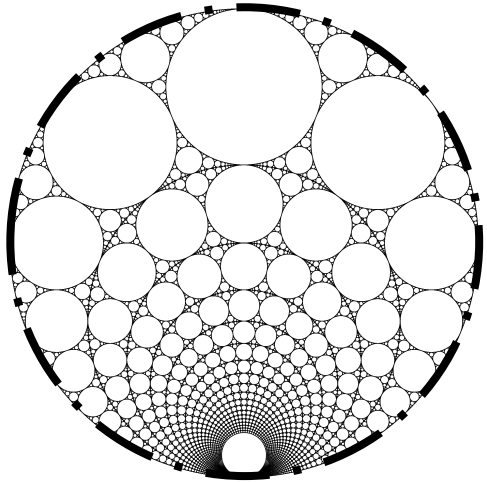
Interoperability: Minimal

The Third Door?

Dimensionality: ?

Interoperability: ?

Ethereum



Dimensionality: Arbitrarily High

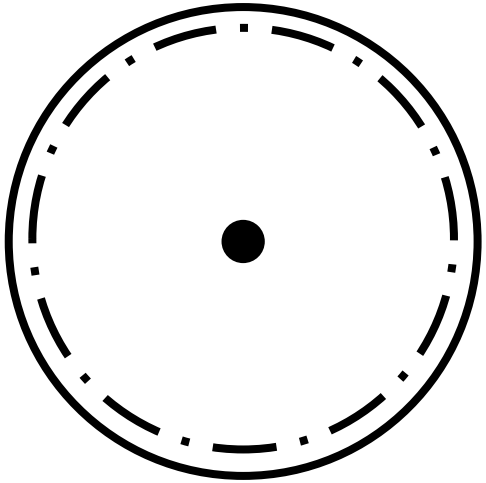
Interoperability: High

Interoperability: The Third Door

- Does there exist a construction which can admit a high degree of interoperability without arbitrary dimensionality?
  - Which allows for a heterogenous spectrum of security assumptions and modes of interoperation?
    - Which doesn’t force applications to bear the externalized incentives of any (all) other(s) they interoperate with, but which also doesn’t preclude them from choosing to do so?
  - Which doesn’t limit the organic discovery of new-uses cases by permissioning or governing the development/deployment of new application behaviors?
  - Which encourages the atomization of security assumptions as an open market process, rather than deferring on the externalities of free or parasitic riders as unobservable?
  - Which limits the contagion risk of individual incentive failures, should they occur?
  - Which can anneal towards an equilibrium in a landscape of jagged tradeoffs?
  - **Which can carry the torch of decentralized applications, should the security of Ethereum-like blockchains fail?**

Cosmos: The Third Door

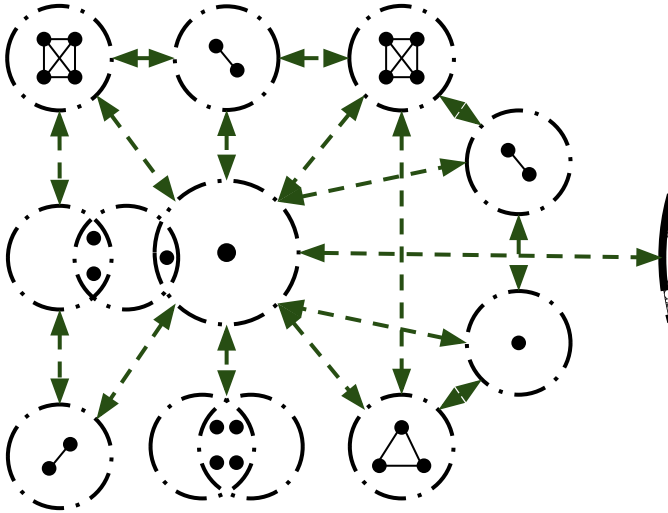
Bitcoin



Dimensionality: Minimal

Interoperability: Minimal

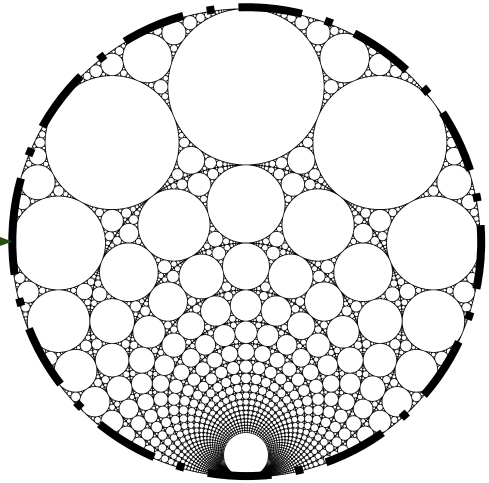
Cosmos



Dimensionality: Finite, Organically Discoverable

Interoperability: Variable, Feasibly High

Ethereum

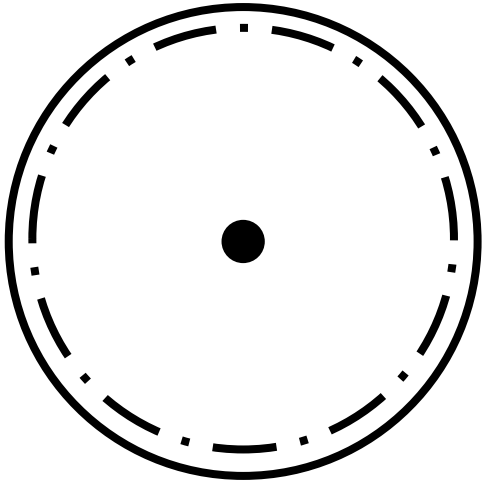


Dimensionality: Arbitrarily High

Interoperability: High

Cosmos: The Third Door

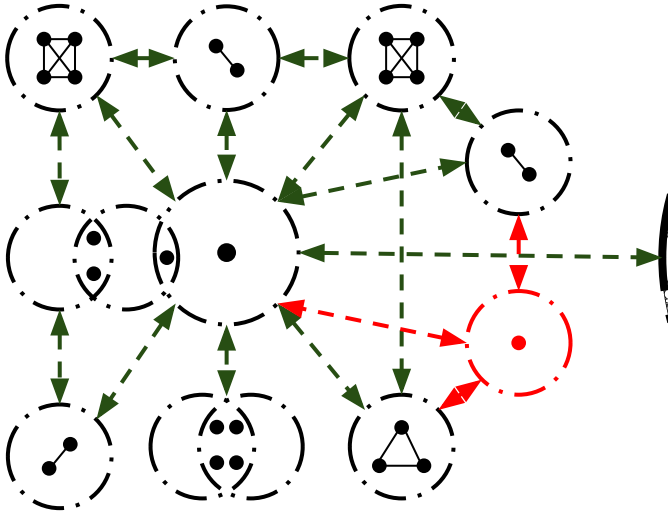
Bitcoin



Dimensionality: Minimal

Interoperability: Minimal

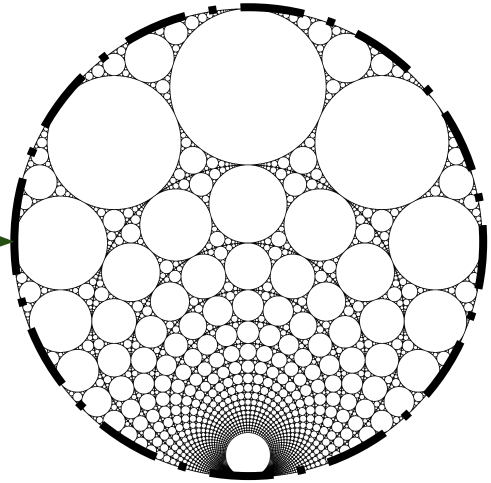
Cosmos



Dimensionality: Finite, Organically Discoverable

Interoperability: Variable, Feasibly High

Ethereum

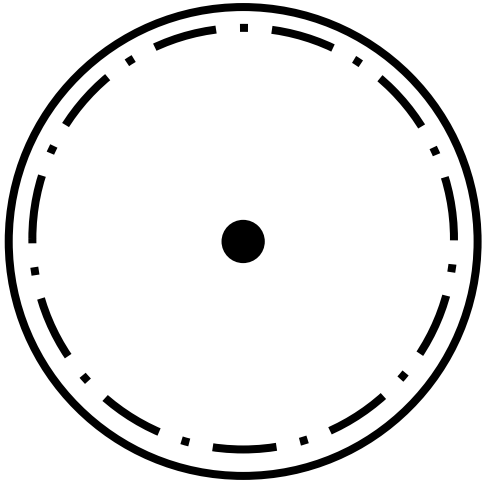


Dimensionality: Arbitrarily High

Interoperability: High

Cosmos: The Third Door

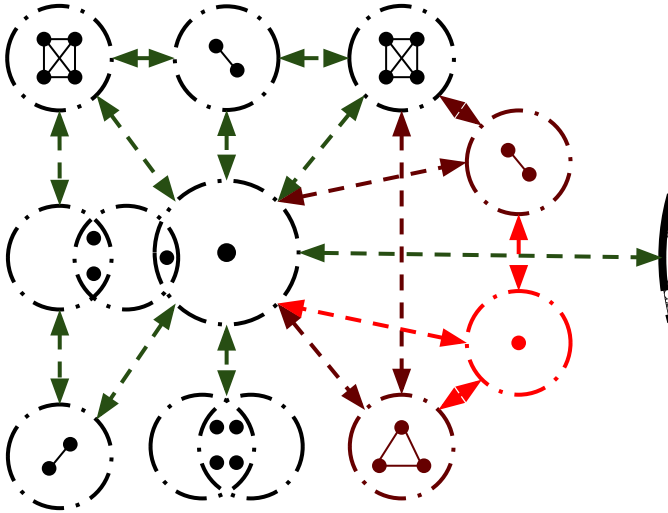
Bitcoin



Dimensionality: Minimal

Interoperability: Minimal

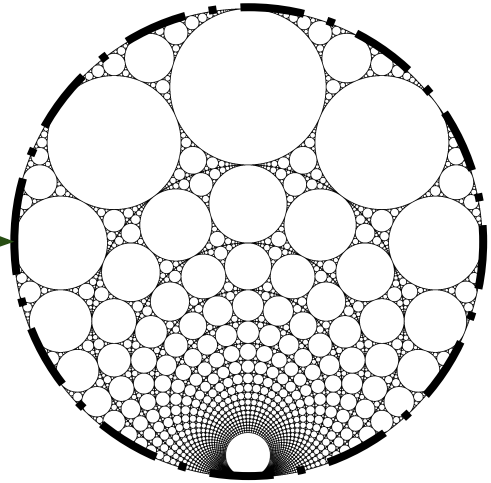
Cosmos



Dimensionality: Finite, Organically Discoverable

Interoperability: Variable, Feasibly High

Ethereum

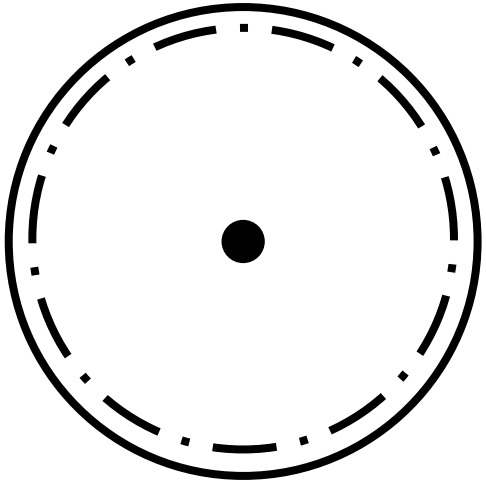


Dimensionality: Arbitrarily High

Interoperability: High

Cosmos: The Third Door

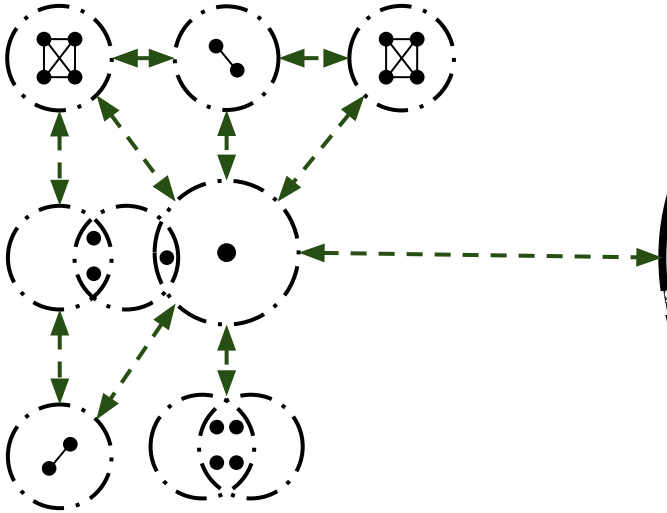
Bitcoin



Dimensionality: Minimal

Interoperability: Minimal

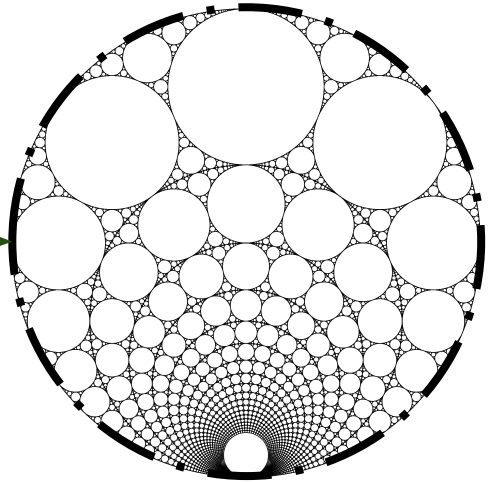
Cosmos



Dimensionality: Finite, Organically Discoverable

Interoperability: Variable, Feasibly High

Ethereum



Dimensionality: Arbitrarily High

Interoperability: High